

Figure 7: Capture filter in Wireshark

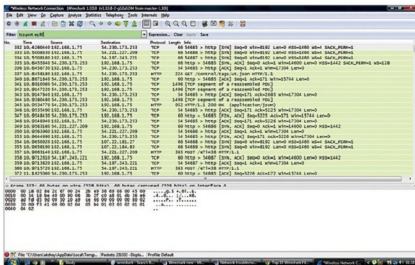


Figure 8: Display filter in Wireshark

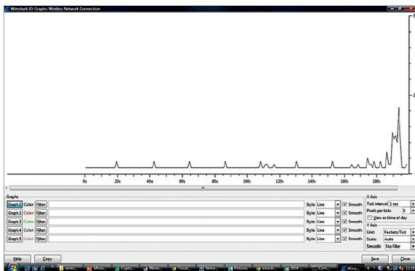


Figure 9: I/O graphs in Wireshark

properties by clicking *Column Preferences*, according to your needs, as shown in Figure 6.

- In Wireshark, we can filter out only the required data from the pool of data using filters, namely, Capture and Display.

The Capture filter is important if you know exactly what kind of data you want. It restricts the basic capturing of non-required packets. Here, we do not have another pool of packets and we capture only the packets that fulfil the filter criteria.

To set the Capture filter, click *Capture → Interfaces → Options* and set the one that you wish to have. We can also use OR and AND like operators for multiple filter options like 'tcp OR udp' as shown in Figure 7.

- Display filters are used if, after sending and capturing a trace, you need to filter out packets based on some conditions (say, protocols or port numbers). For example, we can give 'tcp.port eq 80' to see non-Web data only. We can even run a display filter while taking a trace and also set the filter as 'tcp.port eq 80' as shown in Figure 8.
 - You can save the packets captured simultaneously in various files. Click *Capture → Interfaces → Options*. In the *Capture File* section, specify the file name and location. Click on *multiple files* if you wish to save data captured over a large number of files. You can also initiate to save in the next file, according to size or time.
- The *Ring Buffer* option is used to overwrite existing files. If you give '24' in the *Ring Buffer* section, after 24 files are created, it will overwrite the previous files.
- You can use the telephony features for capturing and monitoring. Click *Telephony → RTP → Show all Streams*. Using this you can monitor every RTP conversation (audio call) within the packet trace.
 - An I/O graphs window (Figure 9) can also be used to see the captured network packets in a graphical format.